

LIHUE, HI, USA

WMO: 911650

Lat: 21.984N

Lon: 159.341W

Elev: 31

StdP: 100.96

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22536

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.9	17.0	10.5	7.9	19.6	11.6	8.5	19.7	12.7	22.0	11.9	21.9	3.2	280	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	30.1	24.2	29.5	23.8	29.0	23.6	25.4	28.3	25.0	28.0	24.6	27.7	6.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	19.6	27.1	23.9	18.9	26.9	23.4	18.3	26.7	77.8	28.5	75.7	28.0	74.1	27.6	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	10.3	9.3	DB	13.8	30.9	1.1	1.0	12.9	31.6	12.3	32.2	11.7	32.7	10.8	33.5	(4)
(5)			WB	11.5	25.7	1.2	1.0	10.7	26.4	10.0	27.0	9.3	27.5	8.4	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	22.2	22.1	22.5	23.5	24.4	25.6	26.2	26.6	26.5	25.8	24.4	23.0	(6)
(7)		DBStd	2.06	1.59	1.44	1.47	1.36	1.26	0.81	0.93	0.95	1.01	1.02	1.30	1.54	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5257	378	338	387	404	447	467	502	514	495	489	432	404	(10)
(11)		CDD18.3	2216	120	104	128	154	188	217	244	256	245	231	182	146	(11)
(12)		CDH23.3	13603	348	248	353	549	992	1472	1972	2274	2154	1763	969	509	(12)
(13)	CDH26.7	1832	8	6	7	18	78	180	329	452	430	245	62	17	(13)	
(14)	Wind	WSAvg	5.8	5.1	5.4	5.7	6.3	5.6	6.2	6.4	5.7	5.6	5.8	5.6	(14)	
(15)	Precipitation	PrecAvg	958	106	90	109	73	64	40	47	49	51	95	115	119	(15)
(16)		PrecMax	1889	362	288	462	271	310	116	175	251	139	458	410	582	(16)
(17)		PrecMin	344	0	0	5	8	2	8	17	0	11	9	2	2	(17)
(18)		PrecStd	322	97	75	95	56	63	25	28	37	30	78	86	113	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	27.5	27.4	27.9	28.8	29.4	30.2	31.1	31.1	29.8	28.8	28.1	(19)
(20)			MCWB	21.3	20.7	21.6	22.4	23.3	23.4	24.1	25.0	24.8	23.9	23.7	22.9	(20)
(21)		2%	DB	26.6	26.3	26.6	27.2	28.1	28.7	29.4	30.1	30.0	29.1	28.0	27.0	(21)
(22)			MCWB	21.1	21.1	21.5	22.1	22.9	23.2	23.8	24.3	24.2	23.6	23.1	22.1	(22)
(23)		5%	DB	25.9	25.6	26.0	26.6	27.6	28.3	28.9	29.4	29.4	28.6	27.4	26.3	(23)
(24)			MCWB	20.9	20.8	21.2	21.8	22.5	23.0	23.6	23.8	23.8	23.4	22.7	21.7	(24)
(25)		10%	DB	25.2	24.9	25.3	26.0	27.0	27.8	28.5	28.9	28.8	28.1	26.8	25.7	(25)
(26)			MCWB	20.7	20.4	20.8	21.5	22.2	22.8	23.4	23.6	23.6	23.2	22.4	21.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.0	23.5	23.7	24.3	24.8	25.7	26.3	26.0	25.3	24.8	24.0	(27)
(28)			MCDWB	25.1	24.9	25.8	26.1	27.5	27.5	28.1	28.8	28.6	28.2	27.3	26.4	(28)
(29)		2%	WB	22.5	22.4	22.5	23.0	23.8	24.2	25.0	25.4	25.3	24.7	24.2	23.2	(29)
(30)			MCDWB	24.9	24.6	25.0	25.8	27.0	27.3	27.9	28.6	28.2	27.7	26.7	25.6	(30)
(31)		5%	WB	21.9	21.8	21.9	22.5	23.1	23.8	24.5	24.8	24.7	24.2	23.6	22.7	(31)
(32)			MCDWB	24.6	24.3	24.6	25.4	26.4	27.0	27.5	28.1	27.9	27.4	26.3	25.2	(32)
(33)		10%	WB	21.4	21.2	21.4	22.0	22.7	23.3	24.0	24.3	24.3	23.8	23.0	22.1	(33)
(34)			MCDWB	24.3	24.0	24.3	25.0	26.0	26.6	27.2	27.7	27.6	27.1	25.8	24.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	5.7	5.6	5.0	5.3	4.8	4.9	4.9	5.2	5.1	4.9	5.3	(35)
(36)			MCDBR	6.6	6.6	6.1	5.2	5.4	5.1	4.9	5.1	5.4	5.2	5.2	5.7	(36)
(37)			MCWBR	3.6	3.7	3.3	2.7	2.6	2.3	2.3	2.3	2.4	2.5	2.8	3.1	(37)
(38)		5% WB	MCDWB	5.9	5.4	5.6	5.0	5.2	4.8	4.7	5.0	5.1	5.1	5.0	5.2	(38)
(39)			MCWBR	3.7	3.5	3.3	2.9	2.7	2.4	2.6	2.5	2.6	2.7	2.9	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.356	0.362	0.394	0.385	0.372	0.373	0.370	0.364	0.367	0.361	0.347	(40)	
(41)		taud	2.608	2.588	2.577	2.473	2.527	2.571	2.577	2.585	2.609	2.615	2.625	2.650	(41)	
(42)		Ebn at Noon	921	937	942	910	908	912	911	919	927	914	906	916	(42)	
(43)		Edh at Noon	89	96	101	113	107	101	101	100	96	92	87	83	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.40	4.92	5.70	6.50	6.97	7.28	7.19	6.90	6.42	5.33	4.48	4.04	(44)	
(45)		RadStd	0.18	0.40	0.61	0.37	0.41	0.19	0.18	0.27	0.19	0.20	0.14	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page